

A Multitask Deep Learning Approach for Sensor-Based Human Activity Recognition and Segmentation

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Abstract—Deep learning (DL) for sensor-based human activity recognition (HAR) has been a focus of research in recent years. Sensor data stream segmentation is a core element in HAR, which has currently been treated as an independent preprocessing task, usually with a fixed-size window. This has led to two critical problems, namely the multiclass window problem caused by possible multiple activities within a fixed-size window and the fluctuation of prediction results due to noisy data and oversegmentation. To address these research challenges, in this article, we conceive a novel multitask DL approach to segmenting and recognizing human activity simultaneously. Specifically, we propose a multi-scale window method based on feature sequence generation to overcome the multiclass window problem. We develop a novel boundary offset prediction algorithm to adjust a window's boundary to tackle the oversegmentation issue. In addition, we design a multitask framework to streamline and optimize the activity recognition and segmentation tasks simultaneously. We conduct extensive experiments on eight benchmark datasets to evaluate the proposed framework and associated methods. Initial results show that our approach outperforms the performance of current state-of-the-art HAR methods.

Index Terms—Activity recognition, activity segmentation, deep learning (DL), multitask learning (MTL), sensors.

NOMENCLATURE

DL	Deep learning.
HAR	Human activity recognition.
HAS	Human activity segmentation.
NED	Normalized edit distance.
IOU	Ratio of the intersection.
NMS	Nonmaximum suppression.

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$L_{\text{conf}}, L_{\text{loc}}$	Classification loss, localization loss.
x, l	Data point, data length.
p	Class probability.
a	Predicted activity.
A_i	Activity class; $a \in \{A_1, A_2, \dots\}$.
s	Window length scale.
$W = (w^x, w^l)$	Window value.
$T = (t^x, t^l)$	Truth boundary value.
$F = (f^x, f^l)$	Boundary offset value.
M	IOU matrix.
α, β	Classification, location weights.

I. INTRODUCTION

SENSOR-BASED HAR is a key technique in many real-world context-aware applications, such as smart homes and healthcare [1], [2], [3]. Early studies [4], [5] showed that although traditional approaches for HAR, such as ontological reasoning [6] and supportive vector machine [7], had achieved satisfactory results, they significantly relied on handcrafted feature engineering and domain knowledge from experts.

In contrast, DL can automatically extract low- and high-level features by training an end-to-end neural network and has been very successful in image classification, video object detection, and natural language processing during the last decade. There has been growing interest in applying DL to HAR in recent years, leading to a large number of studies [8], [9].

Generally speaking, an activity recognition workflow includes sensor data segmentation, feature extraction, and activity classification. Many different models have been explored for feature extraction, including convolution neural network (CNN) [10], selective kernel CNN [11], long short-term memory (LSTM) [12], and self-attention [13]. Nevertheless, Compared with feature extraction, HAS has relatively attracted little attention.

In a real-world application scenario, sensor data streams need to be segmented in HAR systems in real time. The challenge is that it is difficult to define the activity boundaries [14]. Traditional HAS techniques using fixed-size windows have been proved to be ineffective due to the irregularity of the activity duration. As such, selecting a suitable window size is still a challenge. For example, jumping is a short-duration activity and walking may last a long duration. A small

window may lead to misclassification because it contains inadequate information. A large window could contain multiple activities providing more information. Traditional windowing approaches generally take the activity class of the largest number of occurrences as the label of the window [8], [14]. If the window size is not defined appropriately, there could have multiple classes in one window and these activity classes are not identical, leading to the so-called multiclass window problem [15]. The transitions and short periods of activity aggravated this problem, further reducing activity recognition performance.

In recent years, several approaches have been proposed to address the fixed-size window selection and multiclass window challenges. Noor et al. [16] and Akbari et al. [17] proposed an adaptive sliding window method that generates the default size window and then adjusts the size by comparing it with the activity class of the adjacent windows. Guédon [18], Keogh et al. [19], and Fryzlewicz [20] presented change point detection methods. However, these methods are inappropriate in real-time continuous data streams because the boundaries of activities are uncertain. To tackle this problem, Yao et al. [15] exploited the idea of image semantic segmentation for sample-level prediction. This method usually results in some fluctuations in the predicted labels caused by noise or other factors' interference, leading to the so-called oversegmentation problem.

We hypothesize that jointly learning activity segmentation with recognition could be a novel approach to solving multiclass window and oversegmentation problems. Inspired by computer vision object detection, we propose a multiscale window method to tackle the multiclass window problem. Unlike traditional windowing methods, we generate multiscale windows based on feature sequences exploiting the receptive field features. To alleviate the oversegmentation problem, we propose a boundary offset prediction network to predict the activity boundary offset for adjusting the window length and a concatenated algorithm to get the final results of recognition and segmentation. In addition, considering that HAR and HAS share the same feature space, we propose a multitask human activity recognition and segmentation (MTHARS) framework that can provide a mutual constraint on activity recognition and segmentation to streamline the process and optimize the performance of the proposed approach.

The main contributions of this article are as follows.

- 1) We develop the MTHARS multitask framework to streamline and optimize activity recognition and segmentation simultaneously, thus improving both tasks' performances.
- 2) We propose a novel multiscale window-based segmentation method to address the multiclass windows problem.
- 3) For efficient segmentation, we design the activity boundary offset prediction network and a concatenated algorithm to tackle the oversegmentation issue.
- 4) We conduct comprehensive experiments on eight public activity datasets and evaluate the MTHARS method with the state-of-the-art results from relevant studies, demonstrating the effectiveness and superiority of the MTHARS method.

The rest of the article is organized as follows. Section II discusses related works. Section III details the proposed MTHARS approach and associated framework and methods. Section IV presents the experiments and results including discussions on research findings. Section V summarizes our work and discusses future work.

II. RELATED WORKS

A. Deep Learning for Human Activity Recognition

Using DL for sensor-based activity recognition has attracted increasing studies [10], [21]. A CNN for HAR usually consists of convolution, pooling, and fully connected layers [22], [23]. Huang et al. [24] presented a shallower CNN to implement channel information interaction. To better capture the temporal and spatial features, a recurrent neural network (RNN), like gate recurrent unit (GRU) [25] and LSTM [26], was introduced in HAR. Ordóñez and Roggen [27] proposed combining CNN and LSTM to fuse multimodal sensor information for improving activity recognition performance. Guan and Plötz [12] proposed to combine multiple LSTM learners into an ensemble classifier, making it more robust in solving realistic scenario challenges such as noisy, ambiguous data. More recently, attention mechanisms have been investigated for HAR models. Tang et al. [13] presented a ternary attention mechanism with channel, time, and sensor modalities attention. Khan and Ahmad [28] assumed that generating multiple heads with attention can improve feature representations. Abedin et al. [29] introduced a self-attention encoder to learn the latent interactions between multiple sensor channels. Existing models that utilize CNNs for activity recognition typically have the same receptive field size in the neurons of the same layer, which presents a limitation to capture more features [11]. To address this, Gao et al. [11] proposed selection kernel (SK) convolution, which uses the attention mechanism to learn multiscale features by automatically modifying the receptive field.

These existing studies usually use fixed-size window methods for segmentation, suffering from the multiclass window problem. To alleviate this problem, Okeyo et al. [30] proposed a dynamic segmentation model that can shrink and expand the window size by analyzing sensor data, temporal activity information, and the status of the activity recognition. Noor et al. [16] proposed an adaptive sliding window method that compares the window with the activity class of the adjacent windows to adjust the window size. Akbari et al. [17] proposed a hierarchical signal segmentation method, which first applied a larger window to extract features, and then divided the window into smaller windows based on the activity category probabilities and reassigned the labels. Yao et al. [15] developed a full convolution network that uses the dense labeling strategy to assign a class label to each data point. Triboan et al. [31] presented a semantic-based segmentation method inspired by ontological modeling.

B. Multitask Deep Learning for Human Activity Recognition

Multitask learning (MTL) is a learning paradigm, which is built upon the assumption that, for a set of related tasks,

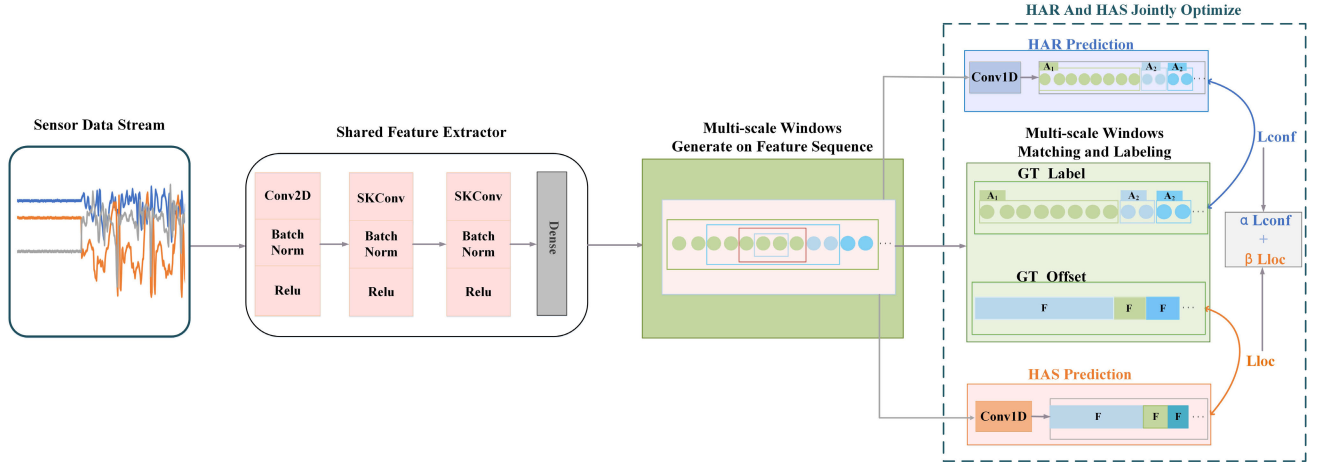


Fig. 1. Overview of the proposed MTHARS framework (GT, Lconf, and Lloc represent ground truth, classification loss, and offset loss, respectively.).

the information contained for each task can help other tasks to improve model generalization. This is usually achieved by learning tasks simultaneously and sharing low-dimensional representation [32].

Ruder [33] summarized two MTL methods for DL: soft or hard parameter sharing of hidden layers. In soft parameter sharing, each task has its own model, but models have similar parameters. In hard parameter sharing, all tasks share hidden layers among them. This can be further classified into three categories [34], namely the multi-input single-output (MISO), the single-input multioutput (SIMO), and the multi-input multioutput (MIMO). It is commonly accepted that HAR and HAS are closely related in the way that both share the same feature space. Based on this observation, the information contained in one task could be helpful to the other, and an integrated approach to jointly performing two tasks could address the challenges associated with HAS and HAR separately in one go. Based on this classification scheme, our framework in this article belongs to SIMO. Zhang and Yang [35] raised three research questions for MTL, i.e., when to share, what to share, and how to share. According to the approach characterization of [35], our method is a Feature Transformation Approach that falls into the category of feature-based.

In the field of DL HAR, there are already studies on MTL, but none of them address simultaneously HAR and HAS. Sun et al. [36], [37] proposed an online MTL framework that was for personalized HAR, where each task corresponds to a specific person in activity recognition. Chen et al. [38] proposed a deep MTL approach, jointly solving activity recognition and user recognition. These studies have demonstrated the advantages of MTL. In this article, we develop a novel MTL framework to optimize activity recognition and segmentation performance simultaneously.

III. MTHARS APPROACH

In this section, we formally define the research problem and describe the MTHARS framework.

A. Problem Definition

The sensor data is denoted by $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i$ denotes a vector of signals collected at time stamp t . $\mathbf{a} = \{a_k\}_{k=1}^K$ is the

set of K activity labels contained in the activity data stream, where $a_k \in \{A_1, A_2, A_3, \dots, A_L\}$, and each a_k has a different duration. Our goal is to segment the streaming sensor data into K nonoverlapping segments $\{\mathbf{X}_k\}_{k=1}^K$, each $\mathbf{X}_k \subseteq \mathbf{X}$, and $\bigcup_{k=1}^K \mathbf{X}_k = \mathbf{X}$. Each $\mathbf{X}_k$ is assigned an a_k . The start and end of $\mathbf{X}_k$ represent an activity boundary, which can also be denoted by a centroid point and corresponding length (t^x, t^l).

B. Holistic Framework of MTHARS

Traditional activity recognition methods usually use a fixed-size window to perform sensor data stream segmentation. This has led to multiclass window and oversegmentation problems. To address them, we propose a multitask framework to jointly optimize the activity recognition and segmentation tasks. The MTHARS framework consists of a selective kernel CNN [11] (as a backbone network), the multiscale window generation module, and the HAR and HAS prediction module, as shown in Fig. 1. It works as follows. First, the backbone network generates the feature sequence. In order to solve the multiclass window problem, the window generation module produces a certain number of multiscale windows. The HAR and HAS prediction module consists of two neural networks in parallel branches. One performs HAR tasks to predict the activity class contained in each window, and the other performs HAS tasks to predict the boundary offset between a window and the truth activity bounding box for addressing the oversegmentation problem. Our recognition and segmentation prediction flow, as shown in Fig. 2, uses an NMS algorithm to retain the best windows. The final results are calculated by Algorithm 1, which concatenates adjacent windows of the same activity.

C. Multiscale Windows Generation on Feature Sequences

1) *Representation of Multiscale Windows*: Unlike the conventional sliding window used in the community of sensor-based HAR, we leverage the concept of anchor in the computer vision field [39]. A window is denoted by (x, l) , where x is the window center and l represents the window length.

2) *Generation of Multiscale Windows*: Assume that the provided feature sequence is n . With a scale of $s \in (0, 1]$, $n\sqrt{s}$ and $n/\sqrt{s}$ represent the window lengths. s takes the values

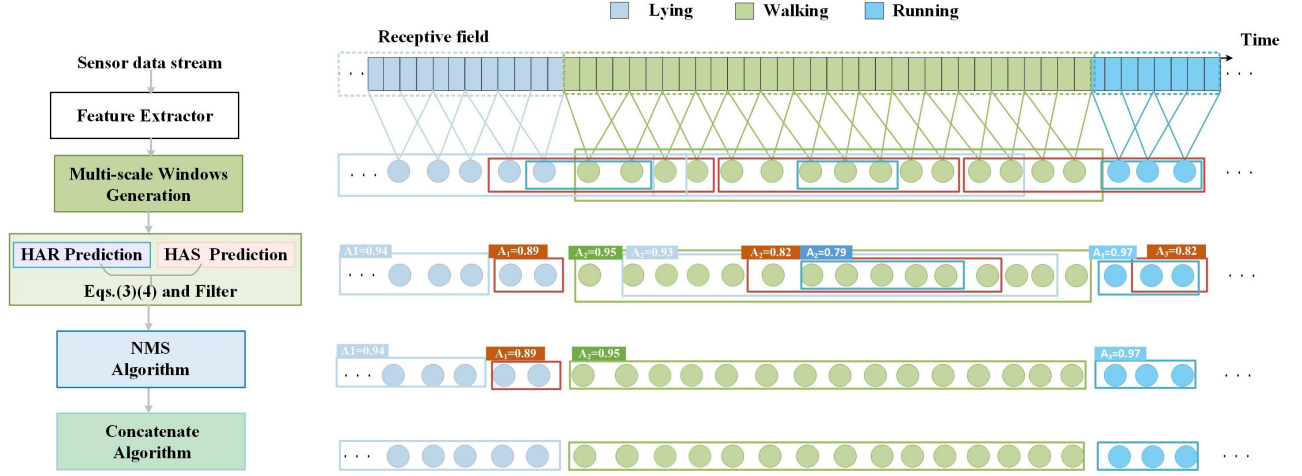


Fig. 2. Overview of the MTHARS's activity recognition and segmentation prediction flow ($A_1 = 0.98$ represents a class probability of 0.98 for activity A_1 in this window).

of $s_1, s_2, \dots, s_m$. This will produce $n \times m \times 2$ windows. Therefore, we can cover various truth activity lengths as much as possible.

We are inspired by the receptive field in convolutional operations to deal with the relationship between sensor data and feature sequences. Therefore, we set the feature sequence length as half of an input data stream length. We generate windows centered on each unit of the feature sequence (as shown in Fig. 2) and divide the center of the window by the feature sequence length. Hence, x indicates the window's relative position in the feature sequence. Since the centers of the window are distributed over all the feature sequence units, these centers are uniformly distributed over the input data stream based on their relative spatial locations.

3) *IOU*: We employ the Jaccard index to measure the similarity between the generated window and the truth activity bounding box. The Jaccard index is defined as the ratio of the intersection length of the window and the truth activity bounding box to their merge length. We denote their Jaccard values as an intersection over union (IOU). The range of an IOU is between 0 and 1, which indicates whether there is any overlap. The value 0 indicates that there is no overlap, whereas 1 indicates that they are equal.

4) *Multiscale Windows Matching and Labeling*: Each generated window is treated as a training sample. To train the model, we need to label each window with an activity class and a bounding offset, where the former refers to the activity associated with the window and the latter refers to the offset of the truth activity bounding box concerning the window.

To mark any generated window, we apply the following approach to assign the truth activity bounding box to the window.

Assume that the generated windows are $W_1, W_2, \dots, W_{na}$ and the truth activity bounding boxes are $T_1, T_2, \dots, T_{nb}$, where $na > nb$. First, we create a matrix $\mathbf{M} \in \mathbb{R}^{(na \times nb)}$, in which the element m_{ij} in the i th row and j th column is the IOU of W_i and T_j . The steps for matching are as follows.

- 1) Find the largest element in $\mathbf{M}$, we set its row index and column index to i_1 and j_1 , respectively, and then assign T_{j_1} to W_{i_1} , discarding all the elements in the i_1 rows and j_1 columns of $\mathbf{M}$.

- 2) Repeat the previous steps until all the elements in column nb of $\mathbf{M}$ have been discarded so that all of the truth activity bounding boxes are allocated to a generated window.
- 3) For each W_i of the remaining $na - nb$ windows, find its best matching T_j , and assign T_j to W_i if the IOU is great than the threshold.

Here is a concrete example. Assume that the largest IOU in $\mathbf{M} \in \mathbb{R}^{(5 \times 3)}$ is m_{51} and T_1 is assigned to W_5 . Then, we discard all the elements in row 5 and column 1 of $\mathbf{M}$

$$\mathbf{M} = \begin{bmatrix} 0.55 & 0.82 & \mathbf{0.96} \\ 0.69 & \mathbf{0.95} & 0.78 \\ 0.32 & 0.48 & 0.88 \\ 0.75 & 0.67 & 0.45 \\ \mathbf{0.98} & 0.67 & 0.88 \end{bmatrix}_{5 \times 3}.$$

Next, we repeat the above steps to find the maximum m_{13} , m_{22} subsequently, and discard row 1 and column 3, and row 2 and column 2. After that, we traverse through the remaining unassigned W_3, W_4 and assign them the truth activity bounding box based on their IOU against the threshold.

Now, we can label a window with an activity class and the boundary offset in the following way. If a window is assigned to a truth activity bounding box, the class of the truth activity bounding box is used to label that window. The boundary offset is calculated as follows.

Given a truth activity bounding box $T = (t^x, t^l)$ and a window $W = (w^x, w^l)$, we define the offset as $F = (f^x, f^l)$. f^x , f^l represents the centre and the length offset, respectively, calculated as follows:

$$f^x = \frac{t^x - w^x}{w^l} \quad (1)$$

$$f^l = \log \frac{t^l}{w^l}. \quad (2)$$

5) *Predicted Activity Boundary*: In the predicting process, we employ the predicted offsets $(\hat{f}^x, \hat{f}^l)$ to calculate the activity boundary $(\hat{t}^x, \hat{t}^l)$, where $\hat{t}^x$ and $\hat{t}^l$ denote the activity

center and the length, respectively,

$$\hat{t}^x = \hat{f}^x w^l + w^x \quad (3)$$

$$\hat{t}^l = w^l \exp(\hat{f}^l). \quad (4)$$

6) *Nonmaximum Suppression*: For the same activity, it may be matched by multiple windows. To find the most suitable window, we modified the NMS algorithm from the computer vision field [40]. This algorithm is described as follows.

The HAR network predicts the probability p for each W . All the predicted windows are ranked as a list L based on p in descending order for the same activity. This list L is then processed as follows.

- 1) Select a W with the highest p as the base and remove the rest of the windows whose IOU with W is over the threshold. Thus, W is retained, and the windows with high similarity to W are discarded. In other words, the window with the nonhighest p is suppressed.
- 2) Repeat the above steps until all windows are selected or nonbase windows are discarded. As such, the IOU values of any two predicted windows are below the threshold.

Algorithm 1 Final Output

Input: Predicted activity-boundary offsets $\mathbf{F} = \{F_i\}_{i=1}^N$

Windows $\mathbf{W} = \{W_i\}_{i=1}^N$

Predicted activity classes $\{a_i\}_{i=1}^N$

Predicted activity probabilities $\{p_i\}_{i=1}^N$

Output: activity boundaries $\hat{\mathbf{T}} = \{\hat{T}_i\}_{i=1}^K$

activity classes $\hat{\mathbf{a}} = \{\hat{a}_i\}_{i=1}^K$

- 1: Remove redundant windows via the NMS algorithm
 - 2: Calculate activity boundaries $\{\mathbf{X}_i\}_{i=1}^L$ with Eqs. (3)(4)
 - 3: $a_1 \leftarrow$ The first activity of $\mathbf{X}_1$
 - 4: $x_1 \leftarrow$ The starting position of the first activity in $\mathbf{X}_1$
 - 5: $x_l \leftarrow$ The ending position of the first activity in $\mathbf{X}_1$
 - 6: Initial index i
 - 7: **for** $j = 1$ to L **do**
 - 8: **if** $a_1 \neq a_j$ **then**
 - 9: Add a_1 to $\hat{a}_i$
 - 10: Add (x_1, x_l) to $\hat{T}_i$
 - 11: $a_1 \leftarrow a_j$
 - 12: $x_1 \leftarrow x_l + 1$
 - 13: $i \leftarrow i + 1$
 - 14: **end if**
 - 15: $x_l \leftarrow$ boundary length of $\mathbf{X}_j + x_l$
 - 16: **end for**
 - 17: **if** $\hat{\mathbf{T}}$ is Empty **then**
 - 18: Add a_1 to $\hat{a}_i$
 - 19: Add (x_1, x_l) to $\hat{T}_i$
 - 20: **end if**
-

D. HAR and HAS Prediction

This component is comprised of two neural networks in parallel branches for HAS and HAR tasks. For the HAR branch, suppose that the number of activity classes is k , and each generated window has $k+1$ classes, where class 0 represents the background. The length of the feature sequence

is n . When centered on each unit of the feature sequence generating m windows, a set of $n \times m \times 2$ windows need to be classified. Choosing a convolutional layer can reduce the number of parameters and does not change the length of the feature sequence. Hence, the output and input coordinates correspond to each other. To generate effective predictions, there are $m(k+1)$ output class channels. For the same spatial position, the output channel with the index $i(k+1) + j$ ($0 \leq j \leq k$) denotes the predictions of class j for the window i .

The design of the HAS branch is similar to the HAR branch. The only difference is that we predict two offsets for each window. We utilize the predicted offset and the absolute position of the window to accurately locate the boundary of the activity. Hence, we are able to address the oversegmentation problem and segment the activity more accurately.

E. Joint Losses

The boundary offset loss (5) is calculated by comparing the truth offset $F = (f^x, f^l)$ with the predicted offset $\hat{F} = (\hat{f}^x, \hat{f}^l)$ using Smooth $_{L_1}$ loss

$$L_{\text{loc}}(F, \hat{F}) = \sum_{i \in \{x, l\}} \text{Smooth}_{L_1}(F_i - \hat{F}_i) \quad (5)$$

in which

$$\text{Smooth}_{L_1}(x) = \begin{cases} 0.5(x)^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \quad (6)$$

The classification loss is calculated by the window labeled activity class $\mathbf{a}$ with the predicted class $\hat{\mathbf{a}}$ using cross-entropy loss (7), in which n represents the sample number

$$L_{\text{conf}}(\mathbf{a}, \hat{\mathbf{a}}) = - \sum_{i=1}^n a_i \log(\hat{a}_i). \quad (7)$$

Some windows not matched with any activity are regarded as negative samples. We sort all the negative samples in descending order according to the activity class probability and select a certain number of negative samples with a higher class probability to participate in the classification loss. The number of negative to positive samples is in the ratio of 3:1. If all the negative samples are involved in the training, it will make the training process tend toward the negative samples.

Finally, we multiply the weight α by the classification loss (conf) plus the weight β by the localization loss (loc)

$$L(a, \hat{a}, F, \hat{F}) = \frac{1}{N} (\alpha L_{\text{conf}}(a, \hat{a}) + \beta L_{\text{loc}}(F, \hat{F})). \quad (8)$$

F. Activity Recognition and Segmentation Prediction

Our prediction process is shown in Fig. 2. After the multi-scale windows generation module, these windows are put into the trained HAR and HAS networks to predict the window categories and the offsets. Next, the length of the windows was adjusted by (3) and (4). Windows with low-class probability are filtered out. Then, we employ an NMS algorithm to remove similar windows. As the length of an activity input to the network is fixed, we input a fixed-length activity data stream at a time to recognize the boundary of each activity. Finally,

TABLE I
SIMPLE DESCRIPTION OF PUBLIC HAR DATASETS

Attribute \ Dataset	SKODA	HCI	PS	WISDM	UCI	OPPORTUNITY	PAMAP2	UNIMIB SHAR
Type	AG	AG	ADL	ADL	ADL	ADL	ADL	ADL
Subject	1	1	4	29	30	4	9	4
Rate	96HZ	96HZ	50HZ	20HZ	50HZ	30HZ	33.3HZ	30HZ
Window Size	1s	1s	2s	10s	2.56s	1s	1s	1s
Activity Categories	10	8	6	6	6	18	18	17
Sample	696975	7352	161959	1,098,208	748406	701366	2872533	11771
Proportion of Training Data	70%	70%	70%	70%	70%	70%	80%	70%
Proportion of Testing Data	30%	30%	30%	30%	30%	30%	20%	30%

¹ AG denotes gesture activity and ADL denotes activity of daily life.

we concatenate all the segments according to the activity class to obtain the boundary of each activity in the whole activity data stream. The concatenation algorithm¹ is described in Algorithm 1.

IV. EXPERIMENTS AND RESULTS

In this section, we detail our extensive experiments on eight benchmark datasets and analyze the experimental results to evaluate the effectiveness and scalability of the proposed MTHARS framework.

A. Datasets

We use eight publicly available benchmark HAR datasets to assess the effectiveness of our framework in daily activity segmentation and recognition. To make a fair comparison with other studies, we select the same parameters. The specifics of the eight benchmark datasets are described in Table I, and the activity length distributions are depicted in Fig. 3.

- 1) *SKODA Dataset [41]*: A participant wearing ten USB acceleration sensors in the left and right hands, respectively, performed ten different gestural activities in an automobile repair scenario. Each of the gesture activities was conducted more than 70 times.
- 2) *HCI Dataset [42]*: Eight accelerometer USB sensors were worn on the participant's right arm to perform various gestures, such as sketching triangles, squares, and circles.
- 3) *PS Dataset [43]*: Four participants recorded their walking, standing, running, sitting, going upstairs, and going downstairs activities. They employed the phone's built-in accelerometer, magnetometer, and gyroscope on different locations: pants pocket, waistband, right arm, and right wrist.
- 4) *WISDM Dataset [44]*: The data were obtained by 29 participants using phones with triaxial acceleration sensors placed in their pant pockets with a sampling frequency of 20 Hz. Walking, strolling, walking upstairs, walking downstairs, standing motionless, and standing up were among the six daily activities undertaken by each participant. The mean value of the column was used to fill in the missing values in the dataset.

- 5) *UCI Dataset [45]*: The dataset was generated by 30 participants aged 19–48 years performing daily activities. Each participant wore a Samsung Galaxy S2 smartphone around their waist. They collected sensor data in nine dimensions using the phone's built-in acceleration, gyroscope, linear acceleration, and three-axis angular velocity sensors. Six different daily activities (walking, walking upstairs, walking downstairs, standing still, standing, and lying down) were performed.
- 6) *OPPORTUNITY Dataset [45]*: IMU sensors were placed on volunteers' 12 different positions. They were asked to repeat a sequence of 17 morning activities in the kitchen, such as opening the refrigerator door, closing the refrigerator door, opening the drawer, and closing the drawer. Interpolation was performed to fill in missing values in the dataset.
- 7) *PAMAP2 Dataset [46]*: Nine participants wore an IMU on their chest, hands, and ankles, collecting three types of sensor information. Each participant was required to complete 12 mandatory activities, such as lying, standing, and walking upstairs and downstairs, as well as six elective activities, such as watching television, driving, and playing ball. During the activity transformation, interference was generated. The start and the last 10 s of each activity were eliminated to reduce the noise data.
- 8) *UNIMIB SHAR Dataset [49]*: The dataset was collected from the University of Milan-Bicocca. Thirty volunteers were equipped with Bosh BMA220 sensor Samsung cell phones. These were placed in their left or right pockets to collect daily life activities (running, going upstairs, standing, walking, and so on) and different falling activities. Each activity was repeated 3–6 times.

B. Experimental Setup

We implement the MTHARS framework in the PyTorch platform and train it in Ubuntu 20.04 environment with an RTX 3090 GPU. We set the training epoch as 500 and use the Adam optimizer with an initial learning rate of 0.001. All datasets are divided into training and testing sets, and all the results are performed on the testing set.

1) *Evaluation Metric*: To evaluate the accuracy of the activity segmentation, we use the NED [50]. NED uses the

¹ <https://github.com/duanfurong/multi-scale-windows>

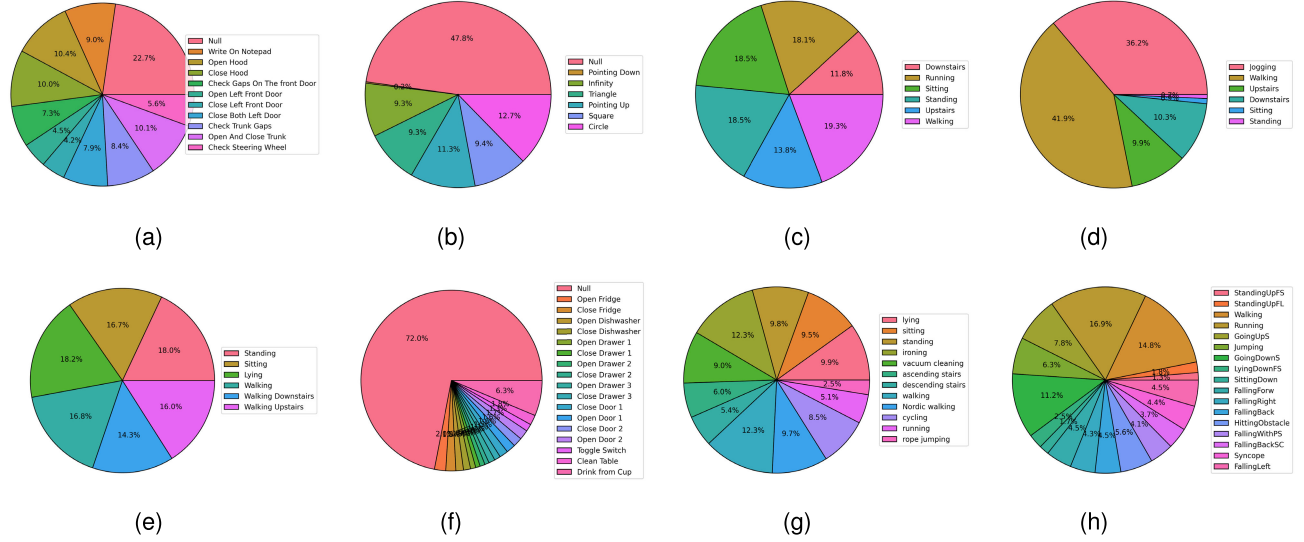


Fig. 3. Activity length distribution of benchmarking datasets. (a) SKODA. (b) HCI. (c) PS. (d) WISDM. (e) UCI. (f) OPPORTUNITY. (g) PAMAP2. (h) UNIMIB SHAR.

TABLE II
OVERALL VERIFICATION TASKS

Sections	Compared Baselines	Tasks
Section V.C	Dynp [18], BottomUP [19], and BinaryCPD [20]	activity segmentation
Section V.D	SK [11], Other methods in [12], [13], [15], [21], [23], [25], [26], [27], [28], [29], [47], [48]	activity recognition

Levenshtein distance to measure the distance between the predicted activity sequence ($\hat{T}$) and the truth activity sequence (T), calculated by the minor operation that makes the two sequences equivalent. NED is defined as follows:

$$\text{NED} = \frac{\text{lev}(\hat{T}, T)}{\text{length of } T} \quad (9)$$

$$\text{lev}(i, j) = \begin{cases} \max(i, j) \min(i, j) = 0 \\ \min = \begin{cases} \text{lev}(i-1, j) + 1 \\ \text{lev}(i, j-1) + 1 \\ \text{lev}(i-1, j-1) + 1_{i \neq j} \end{cases} \end{cases} \quad (10)$$

Equation (10) is the smallest operation step that makes the predicted sequence and the truth sequence equal and considers three different ways to make the two sequences equal, namely, removing an element from the sequence, inserting a new one, and changing the sequence's label directly.

Since the activity classes in human activity data are mostly unbalanced, using classification accuracy is not an appropriate criterion [27]. Therefore, we apply F_1 to evaluate the activity recognition performance

$$F_1 = 2 \sum \frac{N_c}{N_{\text{total}}} \frac{P \times R}{P + R}. \quad (11)$$

Here, N_c denotes the number of class c in all samples and N_{total} the number of all samples. P and R are calculated from

the set of all positive classes, defined below

$$P = \frac{\sum_{i=1}^n \text{TP}_i}{\sum_{i=1}^n \text{TP}_i + \sum_{i=1}^n \text{FP}_i} \quad (12)$$

$$R = \frac{\sum_{i=1}^n \text{TP}_i}{\sum_{i=1}^n \text{TP}_i + \sum_{i=1}^n \text{FN}_i}. \quad (13)$$

Here, i denotes the activity class in the dataset, TP_i the true positive class, FP_i the false-positive class, and FN_i the false-negative class.

2) *Overall Verification Tasks*: We compare various tasks in the MTHARS framework with corresponding tasks in previous studies. Table II describes the verification tasks of MTHARS, including activity segmentation and recognition, respectively.

C. Comparisons With Dynamic Segmentation Algorithms

In order to choose optimal window lengths for HAR and HAS, we adopt dynamic segmentation approaches, aiming to investigate if MTHARS can achieve better segmentation results with the help of classification tasks. Table III displays the segmentation results.

As revealed in Table III, all methods achieve over 90% F_1 values on PS and UCI datasets, although the performance of all approaches on PS is better than that on UCI. The MTHARS framework is superior to other dynamic segmentation approaches. One possible reason is that MTHARS, with

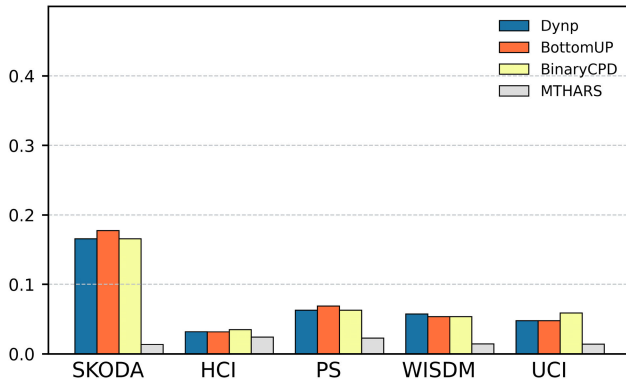


Fig. 4. NED performance on the five benchmark datasets.

TABLE III
 F_1 VALUE OF DYNAMIC SEGMENTATION

Methods \ Datasets	SKODA	HCI	PS	WISDM	UCI
Dynp [18]	0.8858	0.8751	0.9663	0.8711	0.9352
BottomUP [19]	0.8661	0.8750	0.9536	0.8807	0.9353
BinaryCPD [20]	0.8826	0.8450	0.9601	0.8859	0.9151
MTHARS	0.9648	0.9479	0.9733	0.9872	0.9632

the help of the activity recognition task, can improve the segmentation performance.

On the HCI dataset, the NED values of Dynp and BottomUP are similar, and their classification results are similar. On the PS dataset, the NED values of Dynp and BinaryCPD are lower than the BottomUP method, and the F_1 values of their classifications are higher than that of the BottomUP classification. In addition, on the WISDM dataset, the BottomUP and BinaryCPD approaches have higher classification results than Dynp, and their NED values are lower than those of the Dynp method. Especially, We find out that the NED performance on the SKODA dataset is poor due to the fact that the SKODA dataset was collected in an unrestricted environment where each activity lasts for an irregular duration and the number of ground truth segments is relatively small. The degradation caused by this problem is noticeable for sequences with short-time or transition activities. These short-duration activities introduce errors in the testing phase, making it harder for the HAR models to further enhance recognition performance.

As shown in Fig. 4, MTHARS achieves the lowest NED values. At the same time, the activity recognition is more accurate. The reason is that although some activities are short duration, in most cases, they are moderate length, which can reduce the difficulty of activity recognition. Our activity classification results are relatively higher, and our NED values are more stable. This result demonstrates that MTHARS, with the help of the activity recognition task, can improve the performance of segmentation.

To better depict the segmentation results, we visualize the segmentation performance in Fig. 5. We observe that the SKODA dataset contains a lot of short-time activities leading to multiclass window problems. For the WISDM

dataset, where most of the activities are long-duration activities, baseline methods yield oversegmentation problems for short-duration activities. These can be alleviated by using a multiscale window and offset prediction method.

From the above results, we observe that accurate activity segmentation can improve activity recognition performance. In addition, if data segmentation is considered a preprocessing process, errors in data segmentation may be propagated to the later steps. We can combine activity recognition and segmentation to facilitate each other.

D. Comparisons With Activity Recognition Algorithms

In this section, our purpose is to verify that MTHARS can boost the performance of activity recognition accuracy with the help of segmentation tasks. Tables IV and V detail the F_1 value and the accuracy recognition performance, respectively.

Performance Gains: From Tables IV and V, MTHARS exceeds the SK [11] performance with lower model complexity, regardless of which dataset it is based on. For example, on the SKODA dataset, compared with the SK performance, our framework obtains a 0.22% higher F_1 value. Regarding classification accuracy, our framework surpasses the SK by 0.62% with lower model complexity. Compared with SK, MTHARS obtains a 1.47% higher F_1 value on the HCI dataset. On the PS dataset, MTHARS achieves 1.47% F_1 value performance gains with little increase in the computational burden. These comparisons indicate that the features learned by activity recognition and segmentation contain unique information for HAR and can facilitate each other, leading to better performance.

For the WISDM dataset, previous research has found that it is difficult to separate the activities of “DownStairs” and “UpStairs” from Fig. 6, as they have similar patterns. Our proposed framework improves the classification results of both activities. This indicates that the activity classification performance can be better improved with the help of the activity segmentation task. Take the UCI dataset as an example, and among all the activities, the best classification result is “lying” due to the distinct orientation. The activities “sitting” and “standing” have very similar patterns. Therefore, the classification results of both activities are relatively low. MTHARS achieves the best performance among these six activities. The reason is that our framework considers both activity class features and activity boundary features. As shown in Fig. 7, we plot the confusion matrices of MTHARS and SK on the PAMAP2 dataset. It can be observed that our framework improves the performance of the activities of “sitting” and “standing.” These results confirm the advantages of our framework in terms of recognition performance.

In addition, we also compare MTHARS with some state-of-the-art approaches on the HAR datasets, and the experimental results are demonstrated in Table IV. A detailed description of these methods can be found in Section II. It can be concluded that our proposed approach can achieve superior performance.

SK first proposed to adopt the concept of attention to conduct kernel selection among multiple branches with different receptive fields to recognize activity accurately. Compared



Fig. 5. HAS visualization results on SKODA and WISDM datasets. (a) HAS results on the SKODA (samples 1500–2500). (b) HAS results on the WISDM (samples 3000–3400).

TABLE IV
F₁ PERFORMANCE ON EIGHT DATASETS

Methods \ Datasets	SKODA	HCI	PS	WISDM	UCI	OPPORTUNITY	PAMAP2	UNIMIB SHAR
SK [11]	0.9510	0.9377	0.9574	0.9725	0.9558	0.9074	0.9338	0.7463
MTHARS	0.9632	0.9524	0.9721	0.9877	0.9723	0.9213	0.9480	0.7571
Other Reachers	0.924 [12]			0.949 [21]	0.9660 [13]	0.726 [12]	0.854 [12]	0.7538 [13]
	0.916 [15]			0.9263 [26]	0.9293 [23]	0.849 [21]	0.9248 [13]	
	0.958 [27]			0.9720 [28]	0.9302 [47]	0.8058 [48]	0.9116 [23]	
	0.928 [29]				0.9585 [26]	0.915 [27]	0.9303 [48]	
					0.9545 [25]	0.9263 [26]	0.908 [29]	
					0.9537 [28]	0.746 [29]		

TABLE V
ACCURACY AND FLOPS (M) PERFORMANCE ON EIGHT DATASETS

Methods \ Datasets	SKODA	HCI	PS	WISDM	UCI	OPPORTUNITY	PAMAP2	UNIMIB SHAR
SK [11]	0.9586&5.43	0.9341&4.34	0.9649&14.03	0.9751& 17.63	0.9406&8.95	0.9014&10.3	0.9380&45.38	0.7589&6.51
MTHARS	0.9648&4.74	0.9479&3.79	0.9733&12.79	0.9872&6.62	0.9632&8.15	0.9153&9.0	0.9450&38.02	0.7648&5.01

with SK, we are an MTL framework. Regarding the scope of the application, we can more accurately recognize activity and activity boundaries, and we have lower model complexity. We conclude that the MTHARS framework, performing activity segmentation and recognition simultaneously, is able to improve activity recognition performance. The activity segmentation employing multiscale windows and offset prediction can provide an accurate boundary for activity recognition. The two tasks are therefore complementary.

E. Ablation Experiments

In this section, we aim to investigate the effectiveness of our framework. We find that the class loss L_{conf} and offset loss L_{loc} weights, and the scale s of the window are two essential settings.

1) *Impact of Loss Formulation*: Loss formulation is presented in Section III, where L_{conf} and L_{loc} represent the classification loss (7) and the boundary offset loss (5), respectively.

For simplicity, the scale s of the window is fixed to 2 and 3. The detailed results of the experiment are shown in Table VI. We observe that the best result on the WISDM dataset is $2L_{\text{conf}} + 3L_{\text{loc}}$. In addition, the best result is very similar to the result of $L_{\text{conf}} + L_{\text{loc}}$. The results between $L_{\text{conf}} + 2L_{\text{loc}}$ and $2L_{\text{conf}} + L_{\text{loc}}$ are close. These can be attributed to the weights of L_{conf} and L_{loc} being alike. Similarly, on the OPPORTUNITY dataset, we find that the performance of $L_{\text{conf}} + 3L_{\text{loc}}$ is close to $2L_{\text{conf}} + 3L_{\text{loc}}$. Moreover, the $L_{\text{conf}} + L_{\text{loc}}$ brings a 1.53% performance gain compared with the $L_{\text{conf}} + 2L_{\text{loc}}$ on the OPPORTUNITY dataset. Furthermore, we compare the $L_{\text{conf}} + L_{\text{loc}}$ with SK and achieve a noticeable improvement. In summary, we conclude that activity recognition and segmentation can be mutually facilitated. The components of our loss function can promote activity recognition performance.

2) *Impact of Scale s* : Different lengths and numbers of scale s cause different effects. Therefore, we set various combinations of scales s to investigate its performance. Specifically,

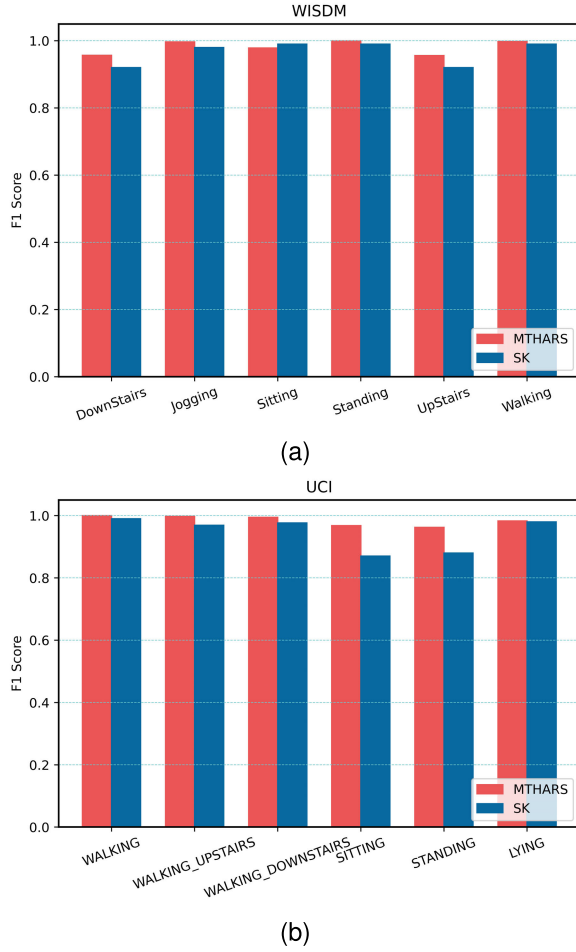


Fig. 6. Comparison of recognition results (F_1 value) for specific classes of human activities. (a) WISDM and (b) UCI.

TABLE VI

ACTIVITY CLASSIFICATION F_1 VALUE WITH DIFFERENT WEIGHT SETTINGS ON BENCHMARK DATASETS

Model	OPPORTUNITY	WISDM
SK [11]	0.9074	0.9725
$L_{conf} + L_{loc}$	0.9213	0.9877
$L_{conf} + 2L_{loc}$	0.9060	0.9796
$L_{conf} + 3L_{loc}$	0.9174	0.9874
$2L_{conf} + L_{loc}$	0.9075	0.9783
$2L_{conf} + 3L_{loc}$	0.9154	0.9881
$3L_{conf} + L_{loc}$	0.9111	0.9841

we set the loss combination as $L_{conf} + L_{loc}$. Due to the dataset length limitation, we only set the following settings, and the detailed results are shown in Table VII. When only set to s , we observe that the performance is inferior to that of several different s . The reason is that the duration of activity is variable, and using a single-size window is insufficient to capture the activity characteristics of different durations. Similarly, setting a small scale on the OPPORTUNITY dataset, such as $s = 0.5$ and 0.3 , may suffer from the same issue. The best classification performance is to set $s = 2, 3$. As most activity duration is irregular, this combination can better enhance activity recognition performance. At the same time, on the UCI dataset, the best result is 0.9723 when set

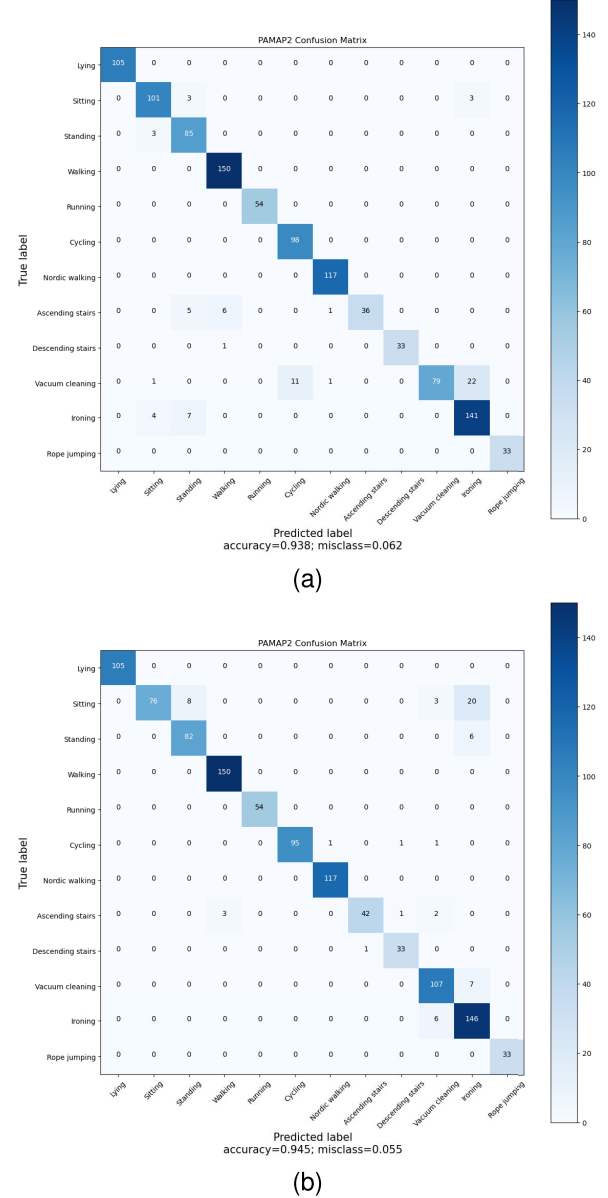


Fig. 7. Confusion matrices on the PAMAP2 dataset between (a) SK [11] and (b) MTHARS.

TABLE VII

ACTIVITY CLASSIFICATION F_1 VALUE FOR DIFFERENT s ON BENCHMARK DATASETS

Model	OPPORTUNITY	UCI
SK [11]	0.9074	0.9558
$s=2$	0.9138	0.9505
$s=0.5, 0.3$	0.9160	0.8928
$s=2, 3$	0.9213	0.9615
$s=2, 3, 4$	0.9167	0.9723

$s = 2, 3, 4$. The second result is 0.9615 with $s = 2, 3$. One possible reason is that most activities are long duration, and therefore a larger window is able to collect more features. We conclude that combining classification and segmentation using multiscale windows can better capture enough activity information to boost the HAR performance.

V. CONCLUSION AND FUTURE WORK

In this article, we have proposed a multitask DL framework for sensor-based human activity recognition and segmentation called MTHARS to jointly improve segmentation and recognition performance. We have developed a multiscale window method to address the multiclass window problem. We have also designed an activity boundary offset prediction network and a concatenated algorithm to tackle the oversegmentation issue. We have conducted comprehensive experiments on a number of publicly available datasets and analyzed the performance of HAR and HAS with the MTHARS against community-recognized state-of-the-art methods. The initial results have demonstrated the effectiveness of the MTHARS. In the future, we plan to address issues related to model complexity so that this new integrated HAR and HAS framework can be deployed on mobile devices and used in real time.

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